DSO 576 | MODULE 6

Project Data Cleaning Assignment

Individual homework. Each student completes and submits their own work.

Clean your project data. You may use Codex, but check its suggestions. Read the rubric before starting.

What to submit

Where	Individual deliverable
Gradescope	One PDF per student. Include the 14-item checklist, cleaning decision log (rule, reason, check), and five record checks (original, expected, actual). Explain how you checked one Codex suggestion. If you did not use Codex, explain how you checked one cleaning decision.
Your GitHub branch	Your cleaning scripts (.py), notebook (.ipynb), data sample, dependency files, and README. In the README, list the data source and version, run instructions, and expected output files.

Submit your own PDF on Gradescope by the posted deadline. There is no group submission.

Your GitHub branch

Push your work to your own branch in the team's project repository. Each student must use a different branch. Example: cleaning-alex123. Do not push this assignment to main or the shared course repository.

Which branch did you push to GitHub? [exact branch name]

Branch URL: [link to your branch] Commit ID: [submitted commit]

Put the branch name, branch URL, and commit ID in both your PDF and README. Give the instructor access to the repository.

How to complete the work

1. Inspect the original data. Write your cleaning rules before applying them.
2. Keep the original files unchanged. Apply your rules and record any changes to them.
3. Check the full dataset. Complete all 14 checklist items with evidence.
4. Restart and run all code. Confirm that it recreates your submitted results.

Include up to 50 cleaned rows, including difficult cases. Explain how you chose them. If the data cannot be shared, provide a clearly labeled synthetic sample and explain how the instructor can review the real results through an approved channel.

Grading rubric

Weight	Criterion	What earns credit
40%	Cleaning decisions	Use rules that fit the project question and the data. Explain exceptions.
40%	Verification	Check the full dataset. Account for changes in counts and totals. Show record checks.
20%	Reproducibility and clarity	Code recreates the submitted results. Instructions and evidence are clear.

Inspect and clean the data

Project: [enter title] Student: [your name]

Fill in the bracketed fields. For each item, choose a status and give evidence. Explain every Not applicable or Unresolved response. Add space as needed.

1 State the project vision and question

State your project vision and one business question. Name the main metric or comparison.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

2 Define one row

What does one row represent? Name the column or combination that should uniquely identify a row, if any.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

3 Preserve the source

List the data source and version or download date. Keep the original files unchanged. Explain how to load them.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

4 Inspect columns and types

List incorrect types and your conversions. Count failed conversions and explain how you handled them.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

5 Standardize text

Show text before and after cleaning. Keep meaningful differences and leading zeros in IDs.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

6 Investigate duplicates

Count exact duplicate rows and repeated IDs separately. Explain which you kept or removed, and why.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

7 Handle missing data

Report missing counts before and after cleaning for each affected column. Explain why you kept, filled, or dropped values.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

Validate the results

Use the same statuses. Give evidence for Done. For Unresolved, explain how the issue affects your analysis.

8 Check suspicious values

Check invalid dates, impossible values, and unusual records. Explain what you corrected, kept, or flagged. Unusual does not always mean wrong.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

9 Verify merges

If you merged tables, list the keys and expected relationship. Report unmatched keys, row counts before and after, and any unexpected extra rows.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

10 Reconcile changes

Report starting and final row counts. Account for the difference. Explain changes to important totals after cleaning or merging.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

11 Check calculations

Explain how missing values affect totals, averages, and denominators. Give the record count for each metric. State how you handle missing group keys.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

12 Verify individual records

Check five records, including difficult cases. Write the expected result and reason before checking the output. Show the original value, actual result, and whether it matches.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

13 Test reproducibility

Restart and run all code from the original data using the listed dependencies. Give the command or notebook steps. Confirm that outputs match your submitted files.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]

14 Document remaining limitations

List unresolved issues. Explain how they could affect your conclusions and what information you still need.

Status: [choose Done / Not applicable / Unresolved]

Evidence or explanation: [result and file or section reference]